

3	C	jumper at rest when $v = 0$, jumper with zero acceleration when gradient of $v - t$ graph is zero.
4	D	Using conservation of momentum $0 = m_{\alpha}v_{\alpha} + m_Xv_X$ giving $\frac{v_{\alpha}}{v_X} = \frac{m_X}{m_{\alpha}} = \frac{3.89 \times 10^{-25}}{6.65 \times 10^{-27}} = 58$
5	D	acceleration of the system = $\frac{F}{m}$